

Large Language Models are Versatile Decomposers: Decomposing Evidence and Questions for Table-based Reasoning

Yunhu Ye^{*†}

University of Science and Technology
of China
yeyunhu@mail.ustc.edu.cn

Binyuan Hui^{*}

DAMO Academy, Alibaba Group
binyuan.hby@alibaba-inc.com

Min Yang[‡]

SIAT, Chinese Academy of Sciences
min.yang@siat.ac.cn

Binhua Li

DAMO Academy, Alibaba Group
binhua.lbh@alibaba-inc.com

Fei Huang

DAMO Academy, Alibaba Group
f.huang@alibaba-inc.com

Yongbin Li[‡]

DAMO Academy, Alibaba Group
shuide.lyb@alibaba-inc.com

ABSTRACT

Table-based reasoning has shown remarkable progress in a wide range of table-based tasks. It is a challenging task, which requires reasoning over both free-form natural language (NL) questions and (semi-)structured tabular data. However, previous table-based reasoning solutions usually suffer from significant performance degradation on “huge” evidence (tables). In addition, most existing methods struggle to reason over complex questions since the essential information is scattered in different places. To alleviate the above challenges, we exploit large language models (LLMs) as decomposers for effective table-based reasoning, which (i) decompose huge evidence (a huge table) into sub-evidence (a small table) to mitigate the interference of useless information for table reasoning, and (ii) decompose a complex question into simpler sub-questions for text reasoning. First, we use a powerful LLM to decompose the evidence involved in the current question into the sub-evidence that retains the relevant information and excludes the remaining irrelevant information from the “huge” evidence. Second, we propose a novel “parsing-execution-filling” strategy to decompose a complex question into simpler step-by-step sub-questions by generating intermediate SQL queries as a bridge to produce numerical and logical sub-questions with a powerful LLM. Finally, we leverage the decomposed sub-evidence and sub-questions to get the final answer with a few in-context prompting examples. Extensive experiments on three benchmark datasets (TabFact, WikiTableQuestion, and FetaQA) demonstrate that our method achieves significantly better results than competitive baselines for table-based reasoning. Notably, **our method outperforms human performance for the first time on the TabFact dataset**. In addition to impressive overall performance, our method also

has the advantage of interpretability, where the returned results are to some extent tractable with the generated sub-evidence and sub-questions. For reproducibility, we release our source code and data at: <https://github.com/AlibabaResearch/DAMO-ConvAI>.

CCS CONCEPTS

- Computing methodologies → Natural language processing;
- Information systems → Retrieval models and ranking.

KEYWORDS

Table-based reasoning, Large language models, Pre-trained language models

ACM Reference Format:

Yunhu Ye, Binyuan Hui, Min Yang, Binhua Li, Fei Huang, and Yongbin Li. 2023. Large Language Models are Versatile Decomposers: Decomposing Evidence and Questions for Table-based Reasoning. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '23)*, July 23–27, 2023, Taipei, Taiwan. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3539618.3591708>

1 INTRODUCTION

Tabular data can be an important complement to textual data, which is informative and ubiquitous in our daily lives. Reasoning about tabular and textual information is a fundamental problem in natural language understanding (NLU) and information retrieval (IR). It can benefit various downstream applications such as table-based fact verification (FV) [2, 8, 16] and table-based question answering (QA) [10, 18, 33, 36, 52]. As shown in Figure 1, table-based reasoning is challenging since it involves sophisticated textual, numerical, and logical reasoning across both unstructured text and (semi-)structured tables. To address the above challenge, table-based reasoning is conventionally accomplished by synthesizing executable languages (e.g., SQL and SPARQL) to interact with tables [1, 12, 20, 21]. However, these methods ignore the semantics of text chunks inside the table, struggling to effectively model web tables with unnormalized free-form text in table cells.

Recently, table-based pre-trained models [4, 15, 17, 23, 29] have proved to be powerful in enhancing reasoning skills on both textual and tabular data, which benefit from the rich knowledge learned from large-scale crawled or synthesized tabular and textual data. For example, TaPas [17] enhanced the understanding of tubular

^{*}Equal Contribution.

[†]Yunhu Ye is also with Shenzhen Institute of Advanced Technology (SIAT), Chinese Academy of Sciences. This work was done when he was interning at Alibaba.

[‡]Corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SIGIR '23, July 23–27, 2023, Taipei, Taiwan

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-9408-6/23/07...\$15.00
<https://doi.org/10.1145/3539618.3591708>

Evidence						
Date	Visitor	Score	Home	Decision	Attendance	Record
Feb. 2	minnersota	4 - 1	columbus	backstorm	18529	30 - 19 - 3
Feb. 5	detroit	3 - 2	minnersota	backstorm	18568	30 - 19 - 4
Feb. 7	dallas	1 - 0	minnersota	backstorm	18568	30 - 20 - 4
Feb. 9	my islanders	3 - 4	dallas	backstorm	18568	31 - 20 - 4

Question During the 2007 - 08 minnesota wild season , minnesota played at home more times than they played away.

Answer True

Figure 1: An example of table-based reasoning.

data by recovering masked cell information in tables. Nevertheless, these models generally need to be fine-tuned on a significant amount of task-specific downstream datasets, struggling to achieve excellent performances when dealing with new datasets with unseen reasoning types. In addition, fine-tuning a pre-trained model on specific tasks generally destroys its in-context ability [42]. Instead of fine-tuning the pre-trained models, in-context learning in large language models (LLMs) [3, 25, 32, 43] has recently gained noticeable attention to exploring the reasoning ability of LLMs, where several input-output exemplars are provided for prompting. For example, the *chain of thought* prompting [43] has been discovered to empower LLMs to perform complex reasoning by generating intermediate reasoning steps.

The prior studies [25, 43] revealed that LLMs could achieve impressive performance on many text reasoning tasks without task-specific model designs and training data. However, the capability of LLMs on table-based reasoning tasks is still under-explored. There are several technical challenges in leveraging LLMs for table-based reasoning with huge tables and complex questions. **First**, since a table could potentially contain a large number of rows and columns, directly encoding all table content via pre-trained models could be computationally intractable and interfere with huge irrelevant information. As pointed out in [6], the LLMs are unable to generalize to “huge” tables with 30+ rows due to the token limitation. Although previous studies have leveraged some approaches [8, 45] such as text matching to retrieve sub-evidence, these approaches usually require a large amount of domain-specific data for training, as the sub-evidence retrieval process requires deep understanding and reasoning of questions and tables. **Second**, decomposing a complex question into simpler sub-questions can effectively facilitate multi-step reasoning of a large model [7, 13, 19]. However, directly decomposing a complex question by leveraging chain-of-thought prompting [43] could easily fall into a hallucination dilemma [22], where the models may generate misleading sub-questions containing information that is inconsistent with the evidence. Chen *et al.* [6] showed that LLMs could sometimes make simple mistakes when performing symbolic operations. This will affect the process of subsequent reasoning, thus we need a more reliable table-based method for complex question decomposition.

To mitigate the aforementioned challenges, in this paper, we explore in-context learning in LLMs to Decompose evidence And

questions for effective Table-based Reasoning (**Dater**). First, we exploit a powerful LLM to decompose the (semi-)structured evidence (a huge table) involved in the current question into relevant sub-evidence (a small table). We implement sub-evidence extraction by predicting the indexes of the rows and columns with the help of a powerful LLM and a few prompts. In this way, we can retain the relevant evidence and exclude the remaining irrelevant evidence from interfering with the decision. In addition, our method has the advantage of interpretability, where the returned table-based results are tractable. Second, we propose a “*parsing-execution-filling*” strategy which explores the programming language SQL to decompose the complex unstructured natural language (NL) question into the logical and numerical computation. Concretely, we generate an abstract logic sub-question by masking the span of numerical values and then converting the abstract logic into SQL query language that is executed on the evidence for obtaining a reliable sub-question. Finally, we leverage the decomposed sub-evidence and sub-questions to get the final answer with the help of a few in-context prompting examples.

The main contributions of this paper are listed as follows:

- We reduce “huge” evidence (a huge table) into “small” sub-evidence (a small table) by predicting the related indexes of rows and columns of the evidence with the help of a powerful LLM and a few prompting examples. Our evidence decomposition method makes the reasoners focus on the essential sub-evidence related to the given question.
- We propose a novel “parsing-execution-filling” strategy to decompose a complex question into simpler step-by-step sub-questions by generating an intermediate SQL as a bridge to produce numerical and logical sub-questions with the help of a powerful LLM and a few prompting examples. Our question decomposition method has proven to be effective in table-based reasoning without requiring a large amount of annotated training data.
- We conduct extensive experiments on three benchmark datasets belonging to table-based fact verification and table-based question answering tasks. Experimental results demonstrate that our Dater method achieves significantly better results than competitive baselines for table-based reasoning. Particularly, Dater outperforms human performance for the first time on the TabFact dataset.
- In addition to impressive overall performance, our Dater also has the advantage of interpretability, where the returned results are to some extent tractable with the generated sub-evidence and sub-questions.

2 RELATED WORK

2.1 Table-based Reasoning

Table-based reasoning requires reasoning over both free-form natural language (NL) questions and (semi-)structured tables [30, 49]. Traditional methods produce executable languages (e.g., SQL and SPARQL) to access the tabular data [1, 12]. However, these methods cannot capture the semantics of text chunks inside a table and fail to model web tables with free-form text in table cells. Recently, several table-based reasoning benchmarks, such as TabFact [8], WikiTable-Question [36], and FetaQA [33], have been proposed to help learn different types of table-based tasks. The availability of large-scale

training data has significantly enhanced the performance of table-based reasoning with the help of deep learning techniques [2, 34].

In parallel, table pre-training has been proposed to encode both tables and texts, which further improves the performance of table-based reasoning. Inspired by the success of masked language modeling (MLM), TaPas [17] enhanced the understanding of tubular data by recovering masked cell information in the table. TAPEX [29] utilized the BART model to imitate the SQL executor in the pre-training stage to obtain better table reasoning capability. ReasTAP [51] designed pre-training tasks based on the reasoning skills of table-based tasks, injecting reasoning skills via pre-training. PASTA [15] introduced a table-operations aware fact verification approach, which pre-trained LMs to be aware of common table-based operations with sentence-table cloze questions synthesized from WikiTables. Subsequently, [6, 9] explored the ability of LLMs for table-based tasks.

2.2 Large Language Models on Reasoning

Large language models (LLMs) have been shown to confer a range of reasoning abilities, such as arithmetic [26], commonsense [28] and symbolic reasoning [54], as the model parameters are scaled up [3]. Notable, chain-of-thought (CoT) [43] leverages a series of intermediate reasoning steps, achieving better reasoning performance on complex tasks. Based on CoT, a number of advanced improvements have been proposed, including ensemble process [41], iterative optimization [46], and example selection [11]. Notably, ZeroCoT [25] improved reasoning performance by simply adding “Let’s think step by step” before each answer. Fu *et al.* [14] proposed complexity-based prompting can generate more reasoning steps for the chain, and achieve significantly better performance. Zhang *et al.* [50] selected examples of in-context automatically by clustering, without the need for manual writing. Despite the remarkable performance of LLMs in textual reasoning, their reasoning capabilities on tabular tasks are still limited. The two most relevant works to this paper are [9] and [6], but none of them focus on the ability of a powerful LLM to decompose the evidence (tables), and the reliability of the reasoning steps.

2.3 Question Decomposition

Question decomposition is essential to understand complex questions. Early work [24] utilized a suite of decomposition rules for question decomposition based on lexicon-syntactic features. The drawback of rule-based methods is that they need experts to manually design rules, making it difficult to extend the rules to new tasks or domains. In recent years, neural models [39, 47] have been proposed to perform question decomposition in an end-to-end manner. Zhang *et al.* [47] proposed a hierarchical semantic parsing method based on a sequence-to-sequence model, which combined a question decomposer and an information extractor. However, these supervised methods rely on a large amount of annotated training data which is labor-intensive to obtain. In parallel, unsupervised decomposition was proposed to produce sub-questions without strong supervision. For example, Perez *et al.* [37] automatically produced a noisy “pseudo-decomposition” for each complex question, which trained a decomposition model on the crawled data with unsupervised sequence-to-sequence learning. Different from

previous decomposition methods, we explore LLMs as versatile decomposers, which decompose both huge evidence and complex questions for table-based reasoning with the help of a powerful LLM and a few prompting examples.

3 PROBLEM FORMULATION AND NOTATIONS

Each instance in table-based reasoning consists of a table T , a natural language question Q , and an answer A . In particular, each table $T = \{v_{i,j} | i \leq \text{Row}_T, j \leq \text{Col}_T\}$ contains Row_T rows and Col_T columns, with $v_{i,j}$ representing the content in the (i, j) -th cell. A question $Q = \langle q_1, \dots, q_n \rangle$ consists of n tokens. In this paper, we focus on two table-based reasoning tasks, including table-based fact verification (FV) and table-based question answering (QA). For table-based FV, the final answer $A \in \{0, 1\}$ is a boolean value that determines the truth or falsity of the input statement. For table-based QA, the answer is a natural language sequence $A = \langle a_1, \dots, a_n \rangle$ that answers the question described by the input statement.

4 METHOD

4.1 In-context Learning

Given a few examples with instructions about detailed specific tasks, Large Language Models (LLMs) can perform in-context learning without training, which learn from analogies. The powerful ability has been widely verified on nature language tasks, including text classification, semantic parsing, mathematical reasoning, etc. Inspired by the advances of LLMs, we aim to explore whether LLMs could tackle reasoning tasks on structured evidence (i.e., tabular data). Formally, the final answer A_{test} can be obtained by predicting $p(A_{\text{test}} | T_{\text{test}}, Q_{\text{test}}, C)$ with evidence table T_{test} and question Q_{test} . Here, $C = \{C_1, \dots, C_{|C|}\}$ is a small set of in-context prompts from manual writing, where each example $C_i = (T_{\text{prompt}}^i, Q_{\text{prompt}}^i, A_{\text{prompt}}^i)$. We provide the detailed in-context prompt as follows.

Prompt 4.1

Instruction: Judge whether a statement is true or false based on the given table.

Table:

table_caption : south wales derby	
col: competition \	total matches \
row 1: league \	55 \ 6/19 \ 16 \ 20
row 2: fa cup \	2 \ 6/0 \ 27 \ 2
row 3: league cup \	5 \ 6/21 \ 0 \ 3

Question:

there are no cardiff wins that have a draw greater than 26 or smaller than 1.

Answer:

False

Table:

table_caption : dan hardy	
col: res \	opponent \
row 1: win \	amir sadollah \
row 2: loss \	duane ludwig \
row 3: win \	chris lytle \

Question:

dan hardy have a game win in turn 4 and he also play against chris lytle in turn 3.

Answer:

True

Table:

table_caption : 2008 - 09 season	
col: nat \	name \
row 1: sco \	chris burke \
row 2: ivory coast \	lucine cherif \
row 3: sco \	alastair park \

Question:

chris burke , be the only player to have a free transfer fee.

Answer:

False

Based on empirical observations, performing in-context learning directly by the above prompt 4.1 cannot achieve optimal performance. Table-based reasoning is a sophisticated and complex task that requires human-like thinking and a fine-grained understanding process. Recently, chain-of-thought [43] proposed a question

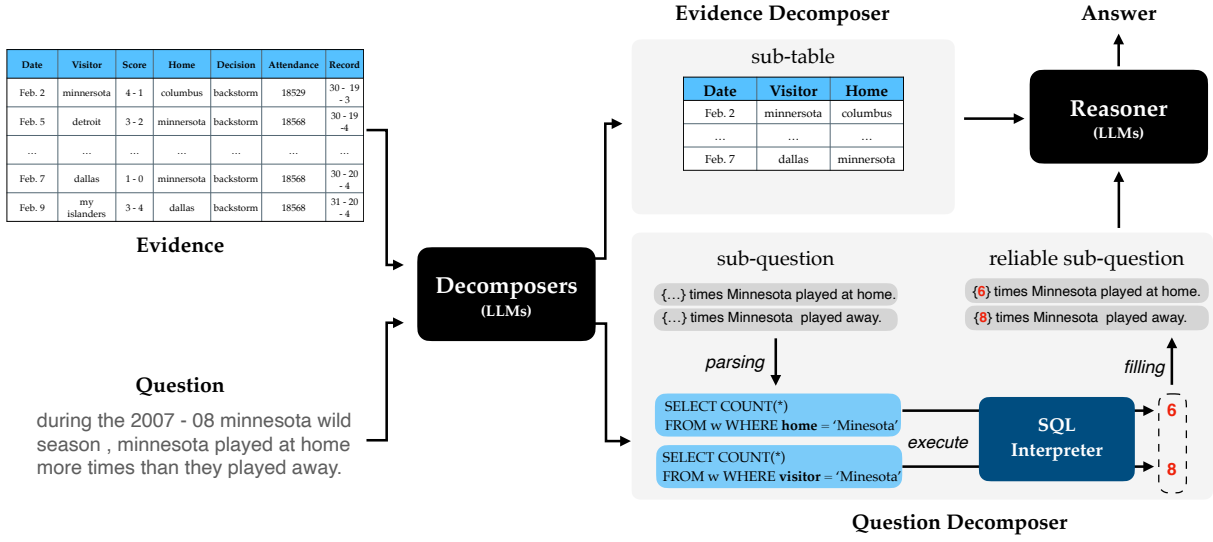


Figure 2: The overview of our Dater framework for table-based reasoning. We first use a powerful LLM (Codex) to probe sub-evidence and sub-questions by performing in-context learning. To obtain a reliable sub-question, we propose a novel “parsing-execution-filling” strategy to alleviate hallucination issues. Ultimately, the reasoner browses through the sub-evidence and sub-questions to get the final answer.

decomposition method, which induces LLMs to perform step-by-step thinking, resulting in better reasoning. However, they did not break down the evidence, resulting in unsatisfactory performance. Specifically, dealing with the ponderous tables directly is usually inefficient and easily interfered with by huge irrelevant information. To this end, we introduce evidence and question decomposition to help large models perform fine-grained reasoning over both evidence (tables) and questions.

4.2 Evidence Decomposer

Humans process table-based reasoning tasks by observing sub-evidence related to the question at hand to complete their judgment. In this paper, we hope to use evidence decomposition to imitate this process. Although previous studies have leveraged some approaches [8, 45] such as text matching to retrieve sub-evidence, empirical results suggest that these approaches are often imperfect and require a large amount of domain-specific data for training, as the sub-evidence retrieval process relies on strong commonsense as well as domain knowledge and requires a joint understanding and reasoning of questions and tables. To this end, we use a powerful LLM to break down the evidence (tables) involved in the current question, retaining the relevant evidence and excluding the remaining irrelevant evidence from interfering with the decision. Concretely, we implement sub-evidence extraction by predicting the indexes of the rows and columns with the help of a powerful LLM and a few prompts. Formally, in the in-context learning stage, the row index $\text{Row}_{\text{test}} = \{\text{Row}_1, \text{Row}_2, \dots, \text{Row}_{|\text{Row}|}\}$ and column index $\text{Col}_{\text{test}} = \{\text{Col}_1, \text{Col}_2, \dots, \text{Col}_{|\text{Col}|}\}$ of sub-evidence table $\hat{T}_{\text{test}}$ can be obtained by predicting $p(\text{Row}_{\text{test}}, \text{Col}_{\text{test}} | T_{\text{test}}, Q_{\text{test}}, C^{\text{ED}})$ with complete evidence T_{test} and corresponding question Q_{test} . $C^{\text{ED}} = \{C_1^{\text{ED}}, \dots, C_{|\text{C}|}^{\text{ED}}\}$ is a small set of in-context examples, where each

C_i^{ED} is the example instance $(\text{Row}_{\text{prompt}}^i, \text{Col}_{\text{prompt}}^i, T_{\text{prompt}}^i, Q_{\text{prompt}}^i)$. Some detailed prompts are illustrated as follows.

Prompt 4.2

Instruction: Select relevant columns / rows in the given table that support or oppose the statement.

Table:

table_caption : south wales derby

col: competition | total matches | cardiff win | draw | swansea win

row 1: league | 55 | 6/19 | 16 | 20

row 2: fa cup | 2 | 6/0 | 27 | 2

row 3: league cup | 5 | 6/21 | 0 | 13

...

Question:

there are no cardiff wins that have a draw greater than 26 or smaller than 1.

Sub-table Column, Row:

col(cardiff win, draw), row(2, 3, 17, 23, 28)

Table:

table_caption : dan hardy

col: res | opponent | event | round

row 1: win | amir sadollah | ufc on fuel tv 4

row 2: loss | duane ludwig | ufc 146 | 1

row 3: win | chris lytle | ufc live 13

...

Question:

Dan hardy have a game win in turn 4 and he also play against chris lytle in turn 3.

Sub-table Column, Row:

col(res, round, opponent), row(1, 3)

Table:

table_caption : 2008 - 09 season

col: nat | name | moving to | type | cost

row 1: sco | chris burke | motherwell | end of contract | free

row 2: ivory coast | lacine cherif | kilmarnock | loan | 7.8 m

row 3: sco | alistair park | hibernian | loan | free

...

Question:

chris burke , be the only player to have a free transfer fee

Sub-table Column, Row:

col(name, cost), row(1, 3, 13)

4.3 Question Decomposer

Decomposing complex questions into step-by-step sub-questions can effectively facilitate the reasoning process of a large model, which has been proven to be effective in numerical and common-sense reasoning [7, 13, 19]. However, we observed that straightforwardly decomposing a complex question by leveraging a chain-of-thought process could easily fall into a hallucination dilemma, i.e., the LLM might not faithfully generate content consistent with the given evidence (tables), especially when numerical values were involved. This will affect the process of subsequent reasoning, thus we need a reliable method for the sub-question generation.

4.3.1 Reliable Sub-questions. To effectively decompose a complex question into step-by-step sub-questions, we propose a “**parsing-execution-filling**” strategy to extend the vanilla chain-of-thought method by exploring the programming language SQL to divide logical steps and numerical computation. Specifically, we first generate an abstract logic sub-question, where we mask the span of numerical values by using cloze style and then convert the abstract logic into SQL queries, similar to text-to-SQL parsing [38, 40]. Afterward, we execute the SQL language on the evidence to get the final result for back-filling, yielding a reliable sub-question. For example, as shown in the bottom of Figure 2, given a question “*during the 2007 - 08 minnesota wild season, minnesota played at home more times than they played away*”, we first mask the spans involving numerical values in sub-questions in prompting examples, and the remaining part can be regarded as the logical question. Here, the logical sub-questions are “*q1: {...} times minnesota played at home.*” and “*q2: {...} times minnesota played away.*”

Formally, a sub-question $\hat{Q}_{\text{test}}$ can be obtained by predicting $p(\hat{Q}_{\text{test}} \mid T_{\text{test}}, Q_{\text{test}}, C^{\text{QD}})$ with the complete evidence T_{test} and complete question Q_{test} . $C^{\text{QD}} = \{C_1^{\text{QD}}, \dots, C_{|C|}^{\text{QD}}\}$ is a small set of in-context prompting examples, where each C_i^{QD} is an example instance $(T_{\text{prompt}}^i, Q_{\text{prompt}}^i, \hat{Q}_{\text{prompt}}^i)$. Detailed prompting examples are provided as follows.

Prompt 4.3		
Instruction: Decompose questions into sub-questions and transform them into a cloze-style format.		
Table: <i>table_caption : south wales derby</i> <i>col: competition total matches cardiff win draw swansea win</i> <i>row 1: league 55 6/19 16 20</i> <i>row 2: fa cup 2 6/0 27 2</i> <i>row 3: league cup 5 6/2 0 3</i>		T_{prompt}^1
Question: there are no cardiff wins that have a draw greater than 26 or smaller than 1.		Q_{prompt}^1
Sub-questions: [1] there are {...} cardiff wins that have a draw greater than 26. [2] there are {...} cardiff wins that have a draw smaller than 1.		$\hat{Q}_{\text{prompt}}^1$
Table: <i>table_caption : dan hardy</i> <i>col: res opponent event round</i> <i>row 1: win amir sadollah ufc on fuel tv 4</i> <i>row 2: loss duane ludwig ufc 146 1</i> <i>row 3: win chris lytle ufc live 3</i>		$T_{\text{prompt}}^{C_1}$
Question: Dan hardy have a game win in turn 4 and he also play against chris lytle in turn 3.		$Q_{\text{prompt}}^{C_1}$
Sub-questions: [1] dan hardy has {...} game win in turn 4. [2] he play {...} game against chris lytle in turn 3.		$\hat{Q}_{\text{prompt}}^{C_1}$
Table: <i>table_caption : 2008 - 09 season</i> <i>col: nat name moving to type cost</i> <i>row 1: sco chris burke motherwell end of contract free</i> <i>row 2: ivory coast lacine cherif kilmarrock loan 7.8 m</i> <i>row 3: sco alistair park hibernian loan free</i>		T_{test}
Question: chris burke , be the only player to have a free transfer fee		Q_{test}
Sub-questions: [1] there are {...} player have a free transfer fee. [2] chris burke is {...} player have a free transfer fee.		$\hat{Q}_{\text{test}}$

The logical statement is then parsed into the SQL queries “SELECT COUNT(*) FROM w WHERE home = ‘minnesota’” and “SELECT COUNT(*) FROM w WHERE visitor = ‘minnesota’”, which are further executed on the evidence, yielding reliable spans “6” and “8” to be backfilled into the placeholder {...} of sub-questions. In this way, we can obtain the reliable sub-questions “*q1: {6} times minnesota played at home.*” and “*q2: {8} times minnesota played away.*”

Formally, the SQL query $\hat{S}_{\text{test}}$ can be obtained by predicting $p(\hat{S}_{\text{test}} \mid T_{\text{test}}, \hat{Q}_{\text{test}}, C^{\text{PR}})$ with the help of complete evidence T_{test} and sub-question $\hat{Q}_{\text{test}}$. Here, $C^{\text{PR}} = \{C_1^{\text{PR}}, \dots, C_{|C|}^{\text{PR}}\}$ is a small set of in-context examples, where each C_i^{PR} is an example instance

$(T_{\text{prompt}}^i, \hat{Q}_{\text{prompt}}^i, \hat{S}_{\text{prompt}}^i)$. Detailed prompting examples are provided as follows.

Prompt 4.4		
Instruction: Generate SQL based on the provided question and table to answer questions.		
Table: <i>table_caption : south wales derby</i> <i>col: competition total matches cardiff win draw swansea win</i> <i>row 1: league 55 6/19 16 20</i>		T_{prompt}^1
Sub-questions: [1] there are {...} cardiff wins that have a draw greater than 26. [2] there are {...} cardiff wins that have a draw smaller than 1.		$\hat{Q}_{\text{prompt}}^1$
Sub-SQL: [1] SELECT COUNT(*) FROM w WHERE draw > 26 [2] SELECT COUNT(*) FROM w WHERE draw < 1		$\hat{S}_{\text{prompt}}^1$
Table: <i>table_caption : dan hardy</i> <i>col: res opponent event round</i> <i>row 1: win amir sadollah ufc on fuel tv 4</i>		$T_{\text{prompt}}^{C_1}$
Sub-questions: [1] dan hardy have {...} game win in turn 4. [2] he play {...} game against chris lytle in turn 3.		$\hat{Q}_{\text{prompt}}^{C_1}$
Sub-SQL: [1] SELECT COUNT(*) FROM w WHERE res == 'win' AND round == 4 [2] SELECT COUNT(*) FROM w WHERE opponent == 'chris lytle' AND round == 3		$\hat{S}_{\text{prompt}}^{C_1}$
Table: <i>table_caption : 2008 - 09 season</i> <i>col: nat name moving to type cost</i> <i>row 1: sco chris burke motherwell end of contract free</i>		T_{test}
Sub-questions: [1] there are {...} player have a free transfer fee. [2] chris burke is {...} player have a free transfer fee.		Q_{test}
Sub-SQL: [1] SELECT COUNT(*) FROM w WHERE cost == 'free' [2] SELECT COUNT(*) FROM w WHERE cost == 'free' AND name == 'chris burke'		$\hat{Q}_{\text{test}}$

4.4 Jointly Reasoning

After performing the above evidence and question decomposition, the reasoner leverages both sub-evidence $\hat{T}_{\text{test}}$ and reliable sub-questions $\hat{Q}_{\text{test}}$ to get a final answer A_{test} by predicting $p(A_{\text{test}} \mid \hat{T}_{\text{test}}, \hat{Q}_{\text{test}}, C^{\text{JR}})$, where $C^{\text{JR}} = \{C_1^{\text{JR}}, \dots, C_{|C|}^{\text{JR}}\}$ is a small set of new in-context prompting examples. Here, each prompting example is denoted as $C_i^{\text{JR}} = (\hat{T}_{\text{prompt}}^i, \hat{Q}_{\text{prompt}}^i, A_{\text{prompt}}^i)$. Detailed prompting examples are provided as follows.

Prompt 4.5		
Instruction: Judge whether a statement is true or false based on the given table.		
Sub-Table: <i>table_caption : south wales derby</i> <i>col: cardiff win draw</i> <i>row 1: 6/0 27</i> <i>row 2: 6/2 0</i>		T_{prompt}^1
Question: there are no cardiff wins that have a draw greater than 26 or smaller than 1.		Q_{prompt}^1
Sub-questions: [1] there are (3) cardiff wins that have a draw greater than 26. [2] there are (2) cardiff wins that have a draw smaller than 1.		$\hat{Q}_{\text{prompt}}^1$
Answer: False		A_{prompt}^1
Sub-Table: <i>table_caption : dan hardy</i> <i>col: res opponent round</i> <i>row 1: win amir sadollah 4</i> <i>row 3: win chris lytle 3</i>		$T_{\text{prompt}}^{C_1}$
Question: dan hardy have a game win in turn 4 and he also play against chris lytle in turn 3.		$Q_{\text{prompt}}^{C_1}$
Sub-questions: [1] dan hardy has (1) game win in turn 4. [2] he plays (1) game against chris lytle in turn 3.		$\hat{Q}_{\text{prompt}}^{C_1}$
Answer: True		$A_{\text{prompt}}^{C_1}$
Sub-Table: <i>table_caption : 2008 - 09 season</i> <i>col: name cost</i> <i>row 1: chris burke free</i> <i>row 3: alistair park free</i>		T_{test}
Question: chris burke , be the only player to have a free transfer fee.		Q_{test}
Sub-questions: [1] there are (3) player have a free transfer fee. [2] chris burke is (1) player have a free transfer fee.		$\hat{Q}_{\text{test}}$
Answer: False		A_{test}

5 EXPERIMENTAL SETUP

5.1 Datasets

We evaluate our proposed method Dater on three table-based reasoning datasets, including TabFact [8], WikiTableQuestion [36], and

Table 1: The hyper-parameters of Codex (code-davinci-002) in the experiments.

Hyper-parameter	TabFact	WikiTableQuestion	FetaQA
Temperature	0.7	0.5	0.5
Top p	1.0	1.0	1.0
Sampling n	20	20	20
Max output tokens	150	150	150
Stop tokens	\n\n	\n\n	\n\n
Num shots	4	2	6

FetaQA [33]. Given the request and cost constraints of the LLMs, we only evaluate our Dater on the test sets of these corpora, without fine-tuning it on the training sets. It is noteworthy that LLMs were trained primarily on the web-crawled and code data. Since the pre-training data of LLMs does not contain tabular data, there is no risk of test data leakage. The details of these three datasets are provided as follows.

- **TabFact** is a table-based fact verification benchmark, which contains statements written by crowd workers based on Wikipedia tables. For example, a *statement*: “*industrial and commercial panel has four more members than the cultural and educational panel.*” is need to be judged whether it is “*True*” or “*False*” according to the given table. We report the accuracy on the test-small set containing 2,024 statements with 298 tables.
- **WikiTableQuestion** contains complex questions annotated by crowd workers based on Wikipedia tables. The crowd workers are asked to write questions that involve several complex operations such as comparison, aggregation and arithmetic operations, which require compositional reasoning over a series of entries in the given table. We use the standard dev set and test set with 2,381 and 4,344 samples, respectively.
- **FetaQA** contains free-form table questions that require deep reasoning and understanding. Most questions in FetaQA are based on information from discontinuous chunks in the table. We evaluate Dater on the test set containing 2,003 samples.

5.2 Evaluation Metrics

TabFact is used to evaluate table-based fact verification which aims to check if a statement is true based on tables. We evaluate binary classification accuracy for the TabFact dataset. For WikiTableQuestion, we use denotation accuracy as our evaluation metric, which verifies whether the predicted answers equal the gold ones. Different from TabFact and WikiTableQuestion datasets that produce short phrase answers, the goal of FetaQA is to generate a complete long-form answer. Therefore, for FetaQA, we utilize BLEU [35], ROUGE-1, ROUGE-2, ROUGE-L [27] as evaluation metrics.

5.3 Implementation Details

In the experiments, we employ GPT-3 Codex (code-davinci-002) as our large language model. For the final reasoning in-context learning step, we annotate 4, 2, 6 prompting examples for TabFact, WikiTableQuestion and FetaQA, respectively. We random select prompting examples and write answers of prompt questions manually. To obtain consistent results, we use a self-consistence decoding strategy [41]. The hyperparameters of Codex for the final reasoning are provided in Table 1.

Table 2: Experimental results on the official small test set of TabFact. Here, “Human” indicates the human performance.

MODEL	TEST
♡ <i>Fine-tuning based Methods</i>	
Table-BERT	68.1
LogicFactChecker	74.3
SAT	75.5
TaPas	83.9
TAPEX	85.9
SaMoE	86.7
PASTA	90.8
w/ Dater	93.0 (↑ 2.2)
Human	92.1
♣ <i>LLM based Methods</i>	
Binder	85.1
Codex	72.6
Dater	85.6 (↑ 13.0)

Table 3: Experimental results on WikiTableQuestion with the official evaluator.

MODEL	DEV	TEST
♡ <i>Fine-tuning based Models</i>		
MAPO	42.7	43.8
MeRL	43.2	44.1
LatentAlignment	43.7	44.5
IterativeSearch	43.1	44.7
T5-3B	-	49.3
TaPas	49.9	50.4
TableFormer	51.3	52.6
TAPEX	58.0	57.2
ReasTAP	58.3	58.6
TaCube	60.9	60.8
OmniTab	61.3	61.2
w/ Dater	62.5 (↑ 1.2)	62.5 (↑ 1.3)
♣ <i>LLM based Methods</i>		
Binder	62.6	61.9
Codex	49.3	47.6
Dater	64.8 (↑ 15.5)	65.9 (↑ 18.3)

5.4 Baselines

We compare the proposed Dater with a range of strong baseline methods that can be divided into two categories: fine-tuning based methods that require training on task-specific data and LLM-based in-context learning methods that do not require fine-tuning.

♡ **Fine-tuning based Methods.** For the fine-tuning based methods, we compare the following methods:

- **Table-BERT** [8] adopts a rule-based method to linearize a table into a natural language (NL) sentence. The linearized table and statement are then directly encoded using a BERT model.
- **LogicFactChecker** [53] employs a sequence-to-action semantic parser to generate a program represented as a tree with multiple

operations. A graph neural network is then utilized to encode statements, tables, and the generated programs.

- **TaPas** [17] enhances the BERT architecture by incorporating the ability to encode tables as an input. It is pre-trained on a joint dataset of text segments and tables collected from Wikipedia.
- **SAT** [48] improves the model’s ability to attend to the relevant information within a table by introducing a structure-aware mask matrix.
- **TAPEX** [29] is designed to make the BART model imitate the SQL executor, which allows the model to acquire better table comprehension ability.
- **SaMoE** [56] introduces mixture-of-experts (MoE) [31] into the field of table-based fact verification, aiming to make different experts focus on different types of reasoning tasks.
- **ReasTAP** [51] defines seven reasoning skills over semi-structured data, including in conjunction, temporal comparison, date difference, and so on.
- **PASTA** [15] designs six types of sentence–table cloze tasks that are pre-trained on a synthesized corpus of 1.2 million items from WikiTables.
- **TableFormer** [44] proposes a method that is robust to perturbations in table rows and columns, thereby improving the model’s understanding of tables through the use of positional encoding.
- **TaCube** [55] improves the ability of PLMs in numerical reasoning by pre-computing aggregation/arithmetic results for the table in advance.
- **OmniTab** [23] proposes an omnivorous pre-training approach that consumes both natural and synthetic data to enhance models with abilities for both types of data.

◆ **LLM based Methods.** For the LLM based methods with in-context learning, we compare the following methods:

- **Codex** [5] directly generates final answer by performing in-context learning as shown in Prompt 4.1.
- **Binder** [9] generates programming language programs and extends the capability of the programming language to solve commonsense problems.

6 EXPERIMENTAL RESULTS

6.1 Main Results

We conduct extensive experiments on TabFact, WikiTableQuestion and FetaQA. The experimental results on TabFact are summarized in Table 2. From the results, we can observe that our Dater model achieves a substantial improvement over the compared methods. If we use LLM (Codex) as reasoner, our Dater achieves an accuracy of 85.6%, which is 13.0% higher than the result predicted by the Codex without evidence and questions decomposition stages. On the other hand, when we use the fine-tuned model as reasoner in Dater framework, i.e., injecting intermediate decomposition generated by Dater into PASTA, the accuracy increases by 2.2% (93.0% vs. 90.8%) over PASTA. It is noteworthy that **Dater outperforms human performance for the first time on the TabFact dataset.**

Table 3 provides the experimental results on the WikiTableQuestion dataset. We observe a similar trend on WikiTableQuestion. Dater achieves a new state-of-the-art accuracy of 65.9% on the test set, surpassing the best baseline method (i.e., Binder) by 4.0%.

Table 4: Experimental results on the FeTaQA test set.

MODEL	BLEU	ROUGE-1	ROUGE-2	ROUGE-L
♡ <i>Fine-tuning based Methods</i>				
T5-small	21.60	0.55	0.33	0.47
T5-base	28.14	0.61	0.39	0.51
T5-large	30.54	0.63	0.41	0.53
◆ <i>LLM based Methods</i>				
Codex	27.96	0.62	0.40	0.52
Dater	30.92	0.66	0.45	0.56

When injecting intermediate decomposition generated by Dater into OmniTab, we can still achieve a gain of 1.3% on the test set. In addition, Dater has an absolute gain of 18.3% compared to the original Codex for table-based QA. The impressive improvement may be due to the fact that WikiTableQuestion has relatively large tables and complex questions, which can be effectively addressed by our table and question decomposition methods.

Table 4 illustrates the experimental results on the FetaQA dataset. We can observe that our Dater method achieves better results than the compared methods T5 and Codex, further validating the effectiveness of Dater.

6.2 Ablation Study

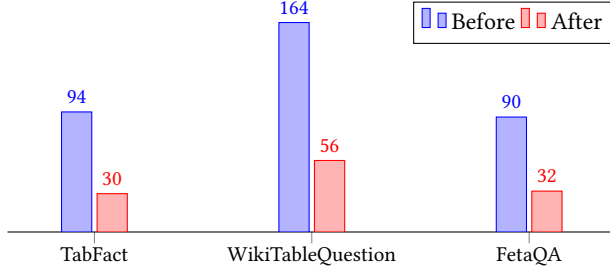
To analyze the impact of two kinds of decomposition in Dater, we conduct an ablation study by discarding the evidence decomposition module (denoted as w/o *Evidence Decomposition*) and the question decomposition module (denoted as w/o *Question Decomposition*) on TabFact, WikiTableQuestion and FetaQA. The questions in the TabFact and WikiTableQuestion datasets can be further divided into two levels based on the difficulty of the questions: *Simple* and *Complex*, which can be used to better evaluate the model performance on different questions. Specifically, TabFact is divided based on officially provided question difficulty labels, while WikiTableQuestion is categorized by the length of the question. While FetaQA questions are more similar in length and difficult to distinguish, so we do not divide them. The ablation test results are reported in Table 5. It is no surprise that combining both evidence and question composition achieves the best performance in all the experimental settings. For TabFact, both evidence decomposition and question decomposition have large impacts on the performance of Dater. The accuracy of Dater without evidence/question decomposition decreases by 3.8%/3.7% on the *All* test set of TabFact. For WikiTableQuestion, question decomposition has a much larger impact than evidence decomposition on the performance of Dater. This is because WikiTableQuestion contains more complex questions that involve comparison, aggregation and arithmetic operations. This verifies the effectiveness of our question decomposition method in dealing with complex questions. For FetaQA, the questions are simpler and involve less numerical reasoning, thus the performance bottleneck is mainly in table content understanding. The performance of Dater drops significantly when discarding evidence decomposition. As FetaQA primarily focuses on the data-to-text generation, it is within our expectation that there is no significant gain from question decomposition.

Table 5: Ablation results on the test sets of the three datasets.

MODEL	TABFACT			WIKITABLEQUESTION			FeTaQA All
	All	Simple	Complex	All	Simple	Complex	
Dater	85.6	91.2	80.0	65.9	68.2	63.5	30.92
w/o <i>Evidence Decomposition</i>	81.8 (↓ 3.8)	86.9 (↓ 4.3)	76.8 (↓ 3.2)	63.9 (↓ 2.0)	65.5 (↓ 2.7)	62.2 (↓ 1.3)	28.46 (↓ 2.46)
w/o <i>Question Decomposition</i>	81.9 (↓ 3.7)	90.0 (↓ 1.2)	74.1 (↓ 5.9)	61.4 (↓ 4.5)	63.6 (↓ 4.6)	59.1 (↓ 4.4)	30.73 (↓ 0.19)

Table 6: Case study on evidence decomposition results predicted by a rule-based method and Dater. We categorize the questions according to the knowledge required for decomposition, including semantic understanding and commonsense knowledge.

SEMANTIC UNDERSTANDING			
Question	on august 25 , remlinger (6 - 6) took the win against the rockies.		
Rule-based method	date, loss		✗
Dater	date, loss, score, opponent		✓
Ground Truth:	date, loss, score, opponent		
COMMONSENSE KNOWLEDGE			
Question	the top scorer (matias suarez) has 5 fewer goals than bart goor , who has 11 goals.		
Rule-based method	player, league		✗
Dater	player, total		✓
Ground Truth:	player, total		
Question	denver did not lose more than one game in a row during november.		
Rule-based method	date		✗
Dater	team, date, score		✓
Ground Truth	team, date, score		

**Figure 3: Comparison of the average number of table cells before and after evidence decomposition on three datasets.**

6.3 Case Study

6.3.1 Case Study for Evidence Decomposition. To better understand how evidence decomposition helps our model capture the relevant sub-evidence, we demonstrate three exemplary cases about sports selected from TabFact. Due to space limitations, we solely provide the table column selection results. From Table 6, we can observe that the rule-based method fails to precisely align the question and evidence since it cannot capture the deep semantic information of both the question and evidence. Taking the first case as an example, the rule-based method cannot link the phrase “against the rockies” and the column name “opponent”. In addition, the rule-based method often suffers from a lack of commonsense knowledge and cannot comprehend complex questions effectively. For example, as shown by the third case, the rule-based method fails to recognize “denver” as a team name and does not discover that the

“score” column can be used to determine the results of the games. On the contrary, our Dater method can address the limitations of previous methods in evidence decomposition and accurately select the relevant columns from a table by eliciting rich semantic and commonsense knowledge from a powerful LLM.

6.3.2 Case Study for Question Decomposition. We use four exemplar cases selected from WikiTableQuestion and TabFact datasets to verify the effectiveness of our question decomposition qualitatively. Table 7 shows the “chain of thought” generated by Codex and our Dater method, where the first two cases are from WikiTableQuestion and the last two cases are from TabFact. From the cases, we can observe that Dater can decompose the complex question into simpler sub-questions that can be easily solved by leveraging a “parsing-execution-filling” strategy. In particular, Dater can generate high-quality SQL queries to retrieve correct information from the evidence. The retrieved information is essential to predict the final answer. In contrast, without question composition, Codex fails to understand and comprehend complex questions, therefore it just gets information from the given question. For example, Codex tends to generate a “chain of thought” (a series of intermediate reasoning steps) that is inconsistent with the evidence.

6.4 Analysis of Sub-evidence

As revealed by [6], the table-based reasoning models are unable to generalize to “huge” tables with 30+ rows, which is the major error source. To address this problem, evidence decomposition is proposed to obtain a sub-evidence by filtering the irrelevant information from the “huge” evidence. To illustrate the effectiveness

Table 7: Case study on question decomposition results predicted by Codex and Dater. We also show the intermediate parsing results, which have the potential to provide the advantage of interpretability.

WikiTableQuestion	
Question 1	what is the number of listings from barrington, farmington, and rochester combined?
Codex CoT	there are totally 1+1+1=3 listings from barrington, farmington, and rochester combined.
Codex Prediction	"3" ✗
Dater Parsing	SELECT COUNT(*) FROM w WHERE city or town = 'barrington' SELECT COUNT(*) FROM w WHERE city or town = 'farmington' SELECT COUNT(*) FROM w WHERE city or town = 'rochester'
Dater CoT	the number of listings from barrington is {1}. the number of listings from farmington is {1}. the number of listings from rochester is {3}.
Dater Prediction	"5" ✓
Gold Answer	"5"
Question 2	does the team have more 5th or 7th place finishes during the regular season?
Codex CoT	the team have 1 5th place finishes and 2 7th place finish during the regular season.
Codex Prediction	"7th" ✗
Dater Parsing	SELECT COUNT(*) FROM w WHERE reg.season LIKE '%5th%' SELECT COUNT(*) FROM w WHERE reg.season LIKE '%7th%'
Dater CoT	the team have {3} 5th place finishes during the regular season. the team have {2} 7th place finishes during the regular season.
Dater Prediction	"5th" ✓
Gold Answer	"5th"
TabFact	
Question 1	pádraig harrington and graeme mcdowell are both from northern ireland.
Codex CoT	pádraig harrington is from northern ireland.graeme mcdowell is from northern ireland.
Codex Prediction	True ✗
Dater Parsing	SELECT COUNT(*) FROM w WHERE player = 'pádraig harrington' AND country = 'northern ireland' SELECT COUNT(*) FROM w WHERE player = 'graeme mcdowell' AND country = 'northern ireland'
Dater CoT	pádraig harrington is {0} from northern ireland. graeme mcdowell is {1} from northern ireland.
Dater Prediction	False ✓
Gold Answer	False
Question 2	atlético ciudad played 28 matches with an average of less than 0.61
Codex CoT	atlético ciudad played 28 matches. the average of atlético ciudad is 0.61.
Codex Predict	True ✗
Dater Parsing	SELECT SUM(matches) FROM w WHERE team = 'atlético ciudad' SELECT COUNT(*) FROM w WHERE team = 'atlético ciudad' AND average < 0.61
Dater CoT	atlético ciudad played {28} matches. atlético ciudad played {0} matches with an average of less than 0.61.
Dater Prediction	False ✓
Gold Answer	False

of evidence decomposition in narrowing the scope of irrelevant evidence, in Figure 3, we report the average number of table cells before and after evidence decomposition on the three datasets. From the results, we can observe that the sub-evidence is about 3x smaller than the original evidence, while achieving better performance. In particular, for WikiTableQuestion, we can reduce the average number of table cells from 164 to 56, significantly relieving the burden of LLMs in dealing with “huge” tables. Similar trends can also be observed on TabFact and FetaQA datasets.

7 CONCLUSION

This paper explored in-context learning in LLMs to decompose structured evidence and unstructured NL questions for effective table-based reasoning. First, we utilized a powerful LLM to decompose the evidence involved in the current question into a relevant sub-evidence. The sub-evidence extraction could identify the relevant sub-evidence and exclude the remaining irrelevant evidence by predicting the indexes of the rows and columns with the help of an LLM and a few prompts. Second, we introduced a “parsing-execution-filling” strategy, which explored the programming language SQL to decompose the complex question into the

logical and numerical computation. Finally, we leveraged the decomposed sub-evidence and sub-questions to obtain the final answer effectively with the help of a few in-context prompting examples. Experimental results on three benchmark datasets demonstrated that our Dater achieved significantly better results than previous competitive baselines (fine-tuning-based methods and LLM-based in-context learning methods) for table-based reasoning. Particularly, Dater outperforms human performance for the first time on the TabFact dataset. In addition to impressive overall performance, Dater also has the advantage of interpretability, where the returned results are tractable with the generated sub-evidence and sub-questions.

8 ACKNOWLEDGEMENTS

Min Yang was supported by National Key Research and Development Program of China (2022YFF0902100), Shenzhen Science and Technology Innovation Program (KQTD20190929172835662), Shenzhen Basic Research Foundation (JCYJ20210324115614039 and JCYJ20200109113441941). This work was supported by Alibaba Group through Alibaba Innovative Research Program.

REFERENCES

- [1] Ibrahim Abdelaziz, Razen Harbi, Zuhair Khayyat, and Panos Kalnis. 2017. A survey and experimental comparison of distributed SPARQL engines for very large RDF data. *Proceedings of the VLDB Endowment* 10, 13 (2017), 2049–2060.
- [2] Rami Aly, Zhijiang Guo, Michael Schlichtkrull, James Thorne, Andreas Vlachos, Christos Christodoulopoulos, Oana Cocarascu, and Arpit Mittal. 2021. Feverous: Fact extraction and verification over unstructured and structured information. *ArXiv preprint abs/2106.05707* (2021). <https://arxiv.org/abs/2106.05707>
- [3] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6–12, 2020, virtual*, Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (Eds.). <https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfc4967418bf8ac142f64a-Abstract.html>
- [4] Zefeng Cai, Xiangyu Li, Binyuan Hui, Min Yang, Bowen Li, Binhua Li, Zhen Cao, Weijie Li, Fei Huang, Luo Si, and Yongbin Li. 2022. STAR: SQL Guided Pre-Training for Context-dependent Text-to-SQL Parsing. In *EMNLP*.
- [5] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. 2021. Evaluating large language models trained on code. *ArXiv preprint abs/2107.03374* (2021). <https://arxiv.org/abs/2107.03374>
- [6] Wenhui Chen. 2022. Large language models are few (1)-shot table reasoners. *ArXiv preprint abs/2210.06710* (2022). <https://arxiv.org/abs/2210.06710>
- [7] Wenhui Chen, Xueguang Ma, Xinyi Wang, and William W Cohen. 2022. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *ArXiv preprint abs/2211.12588* (2022). <https://arxiv.org/abs/2211.12588>
- [8] Wenhui Chen, Hongmin Wang, Jianshu Chen, Yunkai Zhang, Hong Wang, Shiyang Li, Xiyu Zhou, and William Yang Wang. 2020. TabFact: A Large-scale Dataset for Table-based Fact Verification. In *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26–30, 2020*. OpenReview.net. <https://openreview.net/forum?id=rkeJRhNYDH>
- [9] Zhoujun Cheng, Tianbao Xie, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, et al. 2022. Binding language models in symbolic languages. *ArXiv preprint abs/2210.02875* (2022). <https://arxiv.org/abs/2210.02875>
- [10] Minseok Cho, Gyeongbok Lee, and Seung-won Hwang. 2019. Explanatory and Actionable Debugging for Machine Learning: A TableQA Demonstration. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2019, Paris, France, July 21–25, 2019*, Benjamin Piwowarski, Max Chevalier, Éric Gaussier, Yoelle Maarek, Jian-Yun Nie, and Falk Scholer (Eds.). ACM, 1333–1336. <https://doi.org/10.1145/3331184.3331404>
- [11] Antonia Creswell, Murray Shanahan, and Irina Higgins. 2022. Selection-inference: Exploiting large language models for interpretable logical reasoning. *ArXiv preprint abs/2205.09712* (2022). <https://arxiv.org/abs/2205.09712>
- [12] Chris J Date. 1989. *A Guide to the SQL Standard*. Addison-Wesley Longman Publishing Co., Inc.
- [13] Dheeru Dua, Shivanshu Gupta, Sameer Singh, and Matt Gardner. 2022. Successive Prompting for Decomposing Complex Questions. *ArXiv preprint abs/2212.04092* (2022). <https://arxiv.org/abs/2212.04092>
- [14] Yao Fu, Hao Peng, Ashish Sabharwal, Peter Clark, and Tushar Khot. 2022. Complexity-based prompting for multi-step reasoning. *ArXiv preprint abs/2210.00720* (2022). <https://arxiv.org/abs/2210.00720>
- [15] Zihui Gu, Ju Fan, Nan Tang, Preslav Nakov, Xiaoman Zhao, and Xiaoyong Du. 2022. PASTA: Table-Operations Aware Fact Verification via Sentence-Table Cloze Pre-training. *ArXiv preprint abs/2211.02816* (2022). <https://arxiv.org/abs/2211.02816>
- [16] Vivek Gupta, Maitrey Mehta, Pegah Nokhiz, and Vivek Srikumar. 2020. INFOTABS: Inference on Tables as Semi-structured Data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 2309–2324. <https://doi.org/10.18653/v1/2020.acl-main.210>
- [17] Jonathan Herzig, Paweł Krzysztof Nowak, Thomas Müller, Francesco Piccinno, and Julian Eisenschlos. 2020. TaPas: Weakly Supervised Table Parsing via Pre-training. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 4320–4333. <https://doi.org/10.18653/v1/2020.acl-main.398>
- [18] JiaCheng Huang, Wei Hu, Haoxuan Li, and Yuzhong Qu. 2018. Automated Comparative Table Generation for Facilitating Human Intervention in Multi-Entity Resolution. In *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval, SIGIR 2018, Ann Arbor, MI, USA, July 08–12, 2018*, Kevyn Collins-Thompson, Qiaozhu Mei, Brian D. Davison, Yiqun Liu, and Emine Yilmaz (Eds.). ACM, 585–594. <https://doi.org/10.1145/3209978.3210021>
- [19] Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. 2022. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *International Conference on Machine Learning*. PMLR, 9118–9147.
- [20] Binyuan Hui, Ruiying Geng, Qiyu Ren, Binhua Li, Yongbin Li, Jian Sun, Fei Huang, Luo Si, Pengfei Zhu, and Xiaodan Zhu. 2021. Dynamic Hybrid Relation Exploration Network for Cross-Domain Context-Dependent Semantic Parsing. In *AAAI Conference on Artificial Intelligence*.
- [21] Binyuan Hui, Ruiying Geng, Lihan Wang, Bowen Qin, Yanyang Li, Bowen Li, Jian Sun, and Yongbin Li. 2022. S²SQL: Injecting Syntax to Question-Schema Interaction Graph Encoder for Text-to-SQL Parsers. In *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, Dublin, Ireland, 1254–1262. <https://doi.org/10.18653/v1/2022.findings-acl.99>
- [22] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2022. Survey of hallucination in natural language generation. *Comput. Surveys* (2022).
- [23] Zhengbao Jiang, Yi Mao, Pengcheng He, Graham Neubig, and Weizhu Chen. 2022. OmniTab: Pretraining with Natural and Synthetic Data for Few-shot Table-based Question Answering. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Seattle, United States, 932–942. <https://doi.org/10.18653/v1/2022.naacl-main.68>
- [24] Aditya Kalyanpur, Siddharth Patwardhan, BK Boguraev, Adam Lally, and Jennifer Chu-Carroll. 2012. Fact-based question decomposition in DeepQA. *IBM Journal of Research and Development* 56 (2012), 13–1.
- [25] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *ArXiv preprint abs/2205.11916* (2022). <https://arxiv.org/abs/2205.11916>
- [26] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. 2022. Solving quantitative reasoning problems with language models. *ArXiv preprint abs/2206.14858* (2022). <https://arxiv.org/abs/2206.14858>
- [27] Chin-Yew Lin. 2004. ROUGE: A Package for Automatic Evaluation of Summaries. In *Text Summarization Branches Out*. Association for Computational Linguistics, Barcelona, Spain, 74–81. <https://aclanthology.org/W04-1013>
- [28] Jiacheng Liu, Skyler Hallinan, Ximing Lu, Pengfei He, Sean Welleck, Hannaneh Hajishirzi, and Yejin Choi. 2022. Rainier: Reinforced knowledge introspector for commonsense question answering. *ArXiv preprint abs/2210.03078* (2022). <https://arxiv.org/abs/2210.03078>
- [29] Qian Liu, Bei Chen, Jiaqi Guo, Morteza Ziyadi, Zeqi Lin, Weizhu Chen, and Jian-Guang Lou. 2021. Tapex: Table pre-training via learning a neural sql executor. *ArXiv preprint abs/2107.07653* (2021). <https://arxiv.org/abs/2107.07653>
- [30] Zhiqiang Ma, Steven Pomerville, Mingyang Di, and Armineh Nourbakhsh. 2020. SPot: A Tool for Identifying Operating Segments in Financial Tables. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25–30, 2020*, Jimmy Huang, Yi Chang, Xueqi Cheng, Jaap Kamps, Vanessa Murdock, Ji-Rong Wen, and Yiqun Liu (Eds.). ACM, 2157–2160. <https://doi.org/10.1145/3397271.3401406>
- [31] Saeed Masoudnia and Reza Ebrahimpour. 2014. Mixture of experts: a literature survey. *The Artificial Intelligence Review* 42, 2 (2014), 275.
- [32] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2022. Rethinking the Role of Demonstrations: What Makes In-Context Learning Work? *ArXiv preprint abs/2202.12837* (2022). <https://arxiv.org/abs/2202.12837>
- [33] Linyong Nan, Chiachun Hsieh, Ziming Mao, Xi Victoria Lin, Neha Verma, Rui Zhang, Wojciech Kryściński, Hailey Schoelkopf, Riley Kong, Xiangru Tang, Mutethia Mutuma, Ben Rosand, Isabel Trindade, Renusree Bandaru, Jacob Cunningham, Caiming Xiong, Dragomir Radev, and Dragomir Radev. 2022. FeTaQA: Free-form Table Question Answering. *Transactions of the Association for Computational Linguistics* 10 (2022), 35–49. https://doi.org/10.1162/tacl_a_00446
- [34] J. Neeraja, Vivek Gupta, and Vivek Srikumar. 2021. Incorporating External Knowledge to Enhance Tabular Reasoning. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Online, 2799–2809. <https://doi.org/10.18653/v1/2021.naacl-main.224>
- [35] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a Method for Automatic Evaluation of Machine Translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 311–318. <https://doi.org/10.3115/1073083.1073135>
- [36] Panupong Pasupat and Percy Liang. 2015. Compositional Semantic Parsing on Semi-Structured Tables. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, Beijing, China, 1470–1480. <https://doi.org/10.3115/v1/P15-1142>

- [37] Ethan Perez, Patrick Lewis, Wen-tau Yih, Kyunghyun Cho, and Douwe Kiela. 2020. Unsupervised Question Decomposition for Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 8864–8880. <https://doi.org/10.18653/v1/2020.emnlp-main.713>
- [38] Bowen Qin, Binyuan Hui, Lihan Wang, Min Yang, Jinyang Li, Binhua Li, Ruiying Geng, Rongyu Cao, Jian Sun, Luo Si, Fei Huang, and Yongbin Li. 2022. A Survey on Text-to-SQL Parsing: Concepts, Methods, and Future Directions. *ArXiv abs/2208.13629* (2022).
- [39] Alon Talmor and Jonathan Berant. 2018. The Web as a Knowledge-Base for Answering Complex Questions. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. Association for Computational Linguistics, New Orleans, Louisiana, 641–651. <https://doi.org/10.18653/v1/N18-1059>
- [40] Lihan Wang, Bowen Qin, Binyuan Hui, Bowen Li, Min Yang, Bailin Wang, Binhua Li, Fei Huang, Luo Si, and Yongbin Li. 2022. Proton: Probing Schema Linking Information from Pre-trained Language Models for Text-to-SQL Parsing. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (2022).
- [41] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171* (2022).
- [42] Yihan Wang, Si Si, Daliang Li, Michal Lukasik, Felix Yu, Cho-Jui Hsieh, Inderjit S Dhillon, and Sanjiv Kumar. 2022. Preserving In-Context Learning ability in Large Language Model Fine-tuning. *ArXiv preprint abs/2211.00635* (2022). <https://arxiv.org/abs/2211.00635>
- [43] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *ArXiv preprint abs/2201.11903* (2022). <https://arxiv.org/abs/2201.11903>
- [44] Jingfeng Yang, Aditya Gupta, Shyam Upadhyay, Luheng He, Rahul Goel, and Shachi Paul. 2022. TableFormer: Robust Transformer Modeling for Table-Text Encoding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, Dublin, Ireland, 528–537. <https://doi.org/10.18653/v1/2022.acl-long.40>
- [45] Pengcheng Yin, Graham Neubig, Wen-tau Yih, and Sebastian Riedel. 2020. TaBERT: Pretraining for Joint Understanding of Textual and Tabular Data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 8413–8426. <https://doi.org/10.18653/v1/2020.acl-main.745>
- [46] Eric Zelikman, Yuhuai Wu, and Noah D Goodman. 2022. Star: Bootstrapping reasoning with reasoning. *ArXiv preprint abs/2203.14465* (2022). <https://arxiv.org/abs/2203.14465>
- [47] Haoyu Zhang, Jingjing Cai, Jianjun Xu, and Ji Wang. 2019. Complex Question Decomposition for Semantic Parsing. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Florence, Italy, 4477–4486. <https://doi.org/10.18653/v1/P19-1440>
- [48] Hongzhi Zhang, Yingyao Wang, Sirui Wang, Xuezhi Cao, Fuzheng Zhang, and Zhongyuan Wang. 2020. Table Fact Verification with Structure-Aware Transformer. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 1624–1629. <https://doi.org/10.18653/v1/2020.emnlp-main.126>
- [49] Shuo Zhang and Krisztian Balog. 2018. On-the-fly Table Generation. In *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval, SIGIR 2018, Ann Arbor, MI, USA, July 08-12, 2018*, Kevyn Collins-Thompson, Qiaozhu Mei, Brian D. Davison, Yiqun Liu, and Emine Yilmaz (Eds.). ACM, 595–604. <https://doi.org/10.1145/3209978.3209988>
- [50] Zhuosheng Zhang, Aston Zhang, Mu Li, and Alex Smola. 2022. Automatic chain of thought prompting in large language models. *ArXiv preprint abs/2210.03493* (2022). <https://arxiv.org/abs/2210.03493>
- [51] Yilun Zhao, Linyong Nan, Zhenting Qi, Rui Zhang, and Dragomir Radev. 2022. ReasTAP: Injecting Table Reasoning Skills During Pre-training via Synthetic Reasoning Examples. *ArXiv preprint abs/2210.12374* (2022). <https://arxiv.org/abs/2210.12374>
- [52] Victor Zhong, Caiming Xiong, and Richard Socher. 2017. Seq2SQL: Generating Structured Queries from Natural Language using Reinforcement Learning. *CoRR abs/1709.00103* (2017).
- [53] Wanjuan Zhong, Duyu Tang, Zhangyin Feng, Nan Duan, Ming Zhou, Ming Gong, Linjun Shou, Daxin Jiang, Jiahai Wang, and Jian Yin. 2020. Logical-FactChecker: Leveraging Logical Operations for Fact Checking with Graph Module Network. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 6053–6065. <https://doi.org/10.18653/v1/2020.acl-main.539>
- [54] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Olivier Bousquet, Quoc Le, and Ed Chi. 2022. Least-to-most prompting enables complex reasoning in large language models. *ArXiv preprint abs/2205.10625* (2022). <https://arxiv.org/abs/2205.10625>
- [55] Fan Zhou, Mengkang Hu, Haoyu Dong, Zhoujun Cheng, Shi Han, and Dongmei Zhang. 2022. Tacube: Pre-computing data cubes for answering numerical-reasoning questions over tabular data. *ArXiv preprint abs/2205.12682* (2022). <https://arxiv.org/abs/2205.12682>
- [56] Yuxuan Zhou, Xien Liu, Kaiyin Zhou, and Ji Wu. 2022. Table-based Fact Verification with Self-adaptive Mixture of Experts. In *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, Dublin, Ireland, 139–149. <https://doi.org/10.18653/v1/2022.findings-acl.13>